Paul Benedict M. Rosin BSChE - 2B

NAME IT, DRAW IT

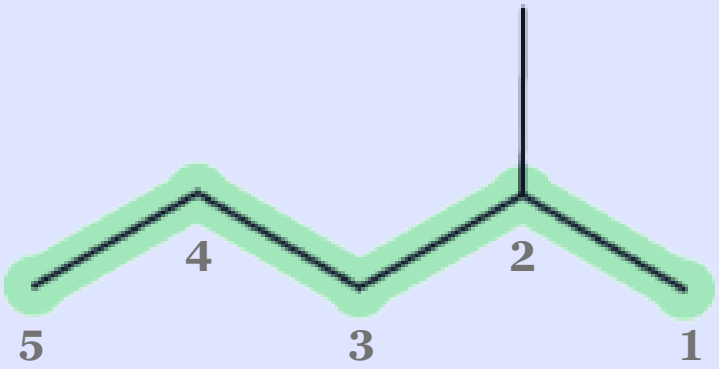
Two-Way IUPAC Nomenclature of Alkanes, Cycloalkanes and Haloalkanes

CHEM 13 — Organic Chemistry (Lecture) | BS Chemical Engineering

CARD A01 - STRUCTURE - NAME

FOUNDATION | MOLECULAR FORMULA: C6H14

1 - NUMBERED PARENT



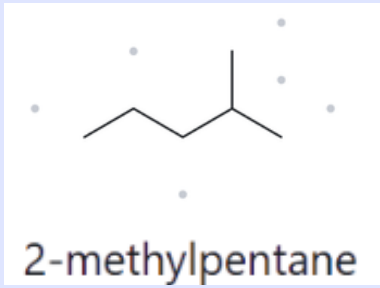
2 - NAMING TRAIL

Parent: longest chain = 5C → **pentane**
Numbering: right → left gives 2; left → right gives 4. 2<4, so start from the right.
Substituents: methyl group on C2 → **2-methyl**

3 - MY NAME (WRITTEN FIRST)

2-methylpentane

4 - THE ALKANE CHECK



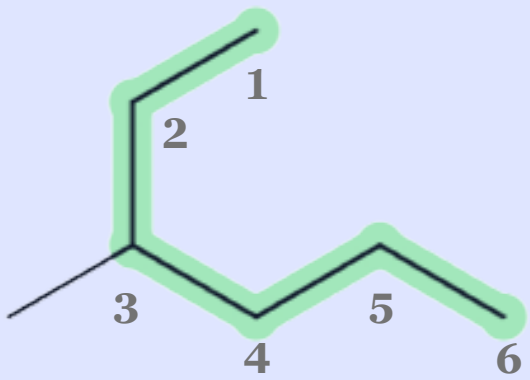
Returned name: 2-methylpentane
✓ Match

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A02 - STRUCTURE - NAME

FOUNDATION | MOLECULAR FORMULA: C7H16

1 - NUMBERED PARENT



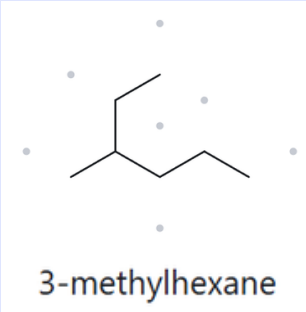
2 - NAMING TRAIL

Parent: longest chain = 6C → **hexane**
Numbering: right → left gives 4; from the uppermost going down gives 3. 3<4, so start from the uppermost.
Substituents: methyl group on C3 → **3-methyl**

3 - MY NAME (WRITTEN FIRST)

3-methylhexane

4 - THE ALKANE CHECK



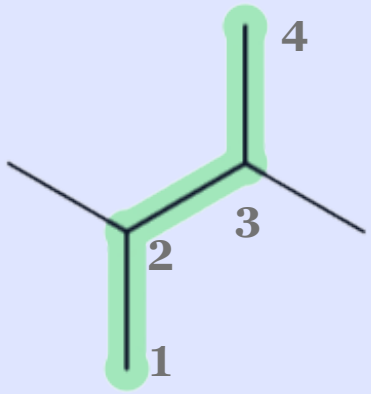
Returned name: 3-methylhexane
✓ Match

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A03 - STRUCTURE - NAME

FOUNDATION | MOLECULAR FORMULA: C6H14

1 - NUMBERED PARENT



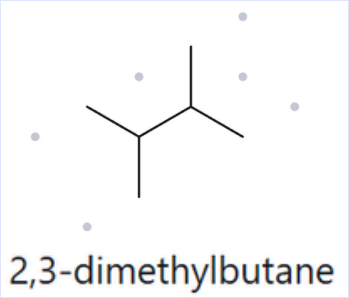
2 - NAMING TRAIL

Parent: longest chain = 4C → **butane**. The same chain can be achieved on 4 other ways.
Numbering: from bottommost to uppermost gives 2 as the lowest locant and the same result can be achieved by starting from the top.
Substituents: 2 methyl on C2 and C3 groups are found→ **2, 3-dimethyl**

3 - MY NAME (WRITTEN FIRST)

2,3-dimethylbutane

4 - THE ALKANE CHECK



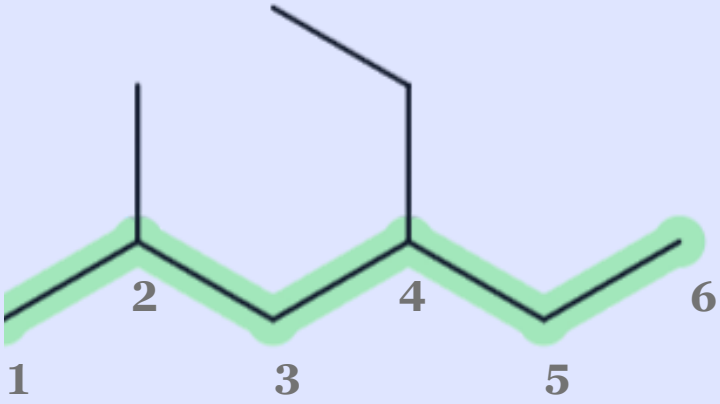
Returned name: 2,3-dimethylbutane
✓ Match

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A04 - STRUCTURE - NAME

DEVELOPING | MOLECULAR FORMULA: C10H22

1 - NUMBERED PARENT



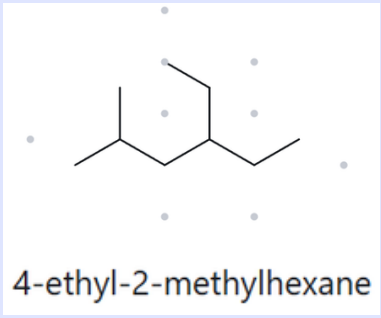
2 - NAMING TRAIL

Parent: longest chain = 6C → **hexane**
Numbering: starting from left to right gives 2 as the lowest locant; starting from right to left gives 3 as the lowest locant. So start from the left.
Substituents: methyl group on C2 and ethyl group on C4 → **4-ethyl-2-methyl**

3 - MY NAME (WRITTEN FIRST)

4-ethyl-2-methylhexane

4 - THE ALKANE CHECK



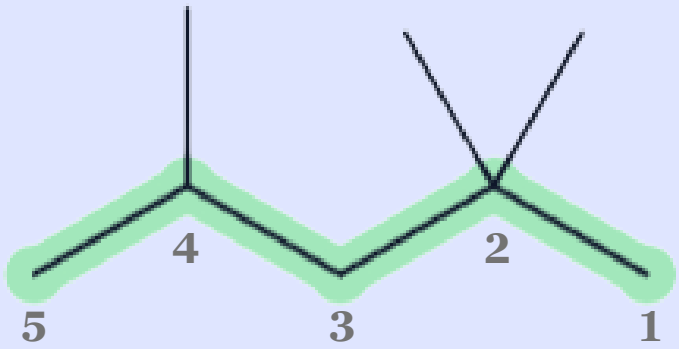
Returned name: 4-ethyl-2-methylhexane
✓ Match

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A05 - STRUCTURE - NAME

DEVELOPING | MOLECULAR FORMULA: C8H18

1 - NUMBERED PARENT



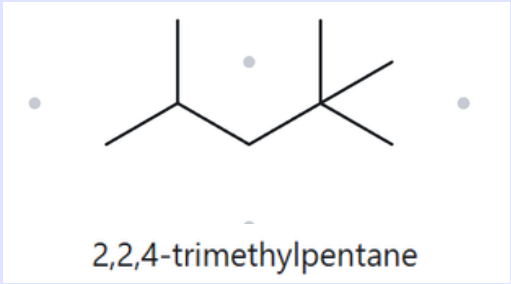
2 - NAMING TRAIL

Parent: longest chain = 5C → **pentane**
Numbering: starting from left to right gives 2 as the lowest locant, same as starting from the right, but starting from left gives 2,4,4 and from right 2,2,4; so start from right.
Substituents: 2 methyl groups on C2 and a methyl group on C4 → **2,2,4-trimethyl**

3 - MY NAME (WRITTEN FIRST)

2,2,4-trimethylpentane

4 - THE ALKANE CHECK



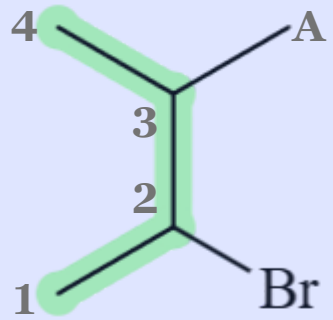
Returned name: 2,2,4-trimethylpentane
✓ Match

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A06 - STRUCTURE - NAME

DEVELOPING | MOLECULAR FORMULA: C5Br

1 - NUMBERED PARENT



2 - NAMING TRAIL

Parent: longest chain = 4C → **butane**. The length can be achieved from CA to C1 but Br is a priority so C1 to C4.

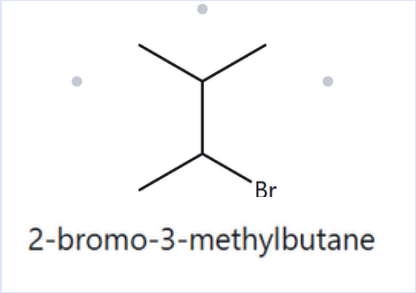
Numbering: starting from the C4 gives 3 for Br while starting from bottom gives 2 as the locant number for Br. So from bottom first.

Substituents: Bromine atom at C2 and methyl group at C3 → **2-bromo-3-methyl**

3 - MY NAME (WRITTEN FIRST)

2-bromo-3-methylbutane

4 - THE ALKANE CHECK



Returned name: 2-bromo-3-methylbutane

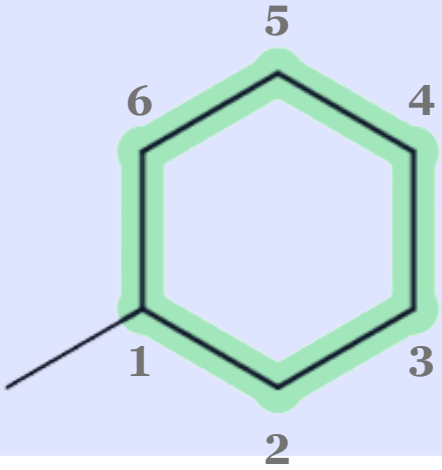
✓ **Match**

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A07 - STRUCTURE - NAME

DEVELOPING | MOLECULAR FORMULA: C7H14

1 - NUMBERED PARENT



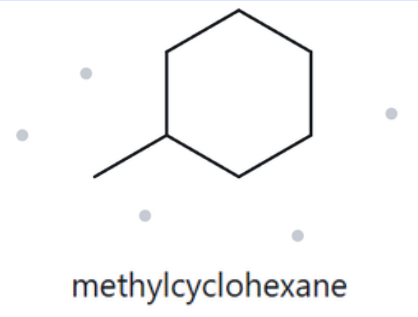
2 - NAMING TRAIL

Parent: ring of 6C is the parent → **cyclohexane**
Numbering: only one substituent present so not needed
Substituents: a methyl group → **methyl**

3 - MY NAME (WRITTEN FIRST)

methylcyclohexane

4 - THE ALKANE CHECK



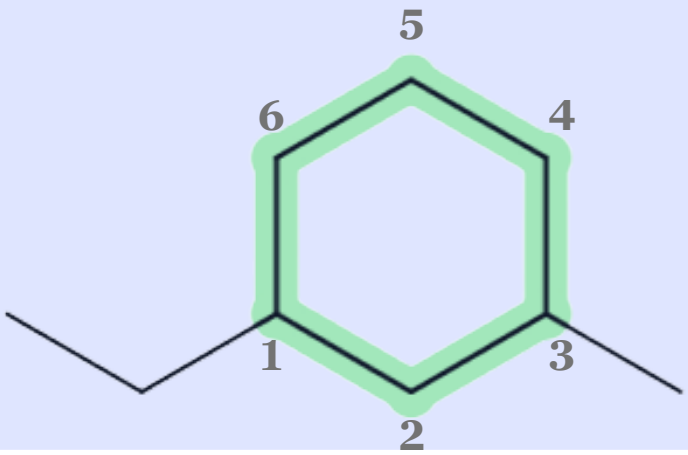
Returned name: methylcyclohexane
✓ **Match**

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A08 - STRUCTURE - NAME

CHALLENGE | MOLECULAR FORMULA: C9H18

1 - NUMBERED PARENT



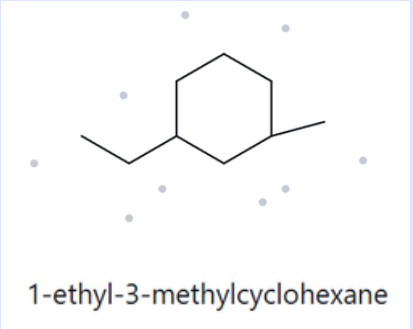
2 - NAMING TRAIL

Parent: ring of 6C is the parent → **cyclohexane**
Numbering: Ethyl group is a priority so this is the start of the chain. Going clockwise gives 5 for methyl while going counterclockwise gives 3. So go counterclockwise at counting.
Substituents: an ethyl group at C1 and a methyl group at C3 → **1-ethyl-3-methyl**

3 - MY NAME (WRITTEN FIRST)

1-ethyl-3-methylcyclohexane

4 - THE ALKANE CHECK



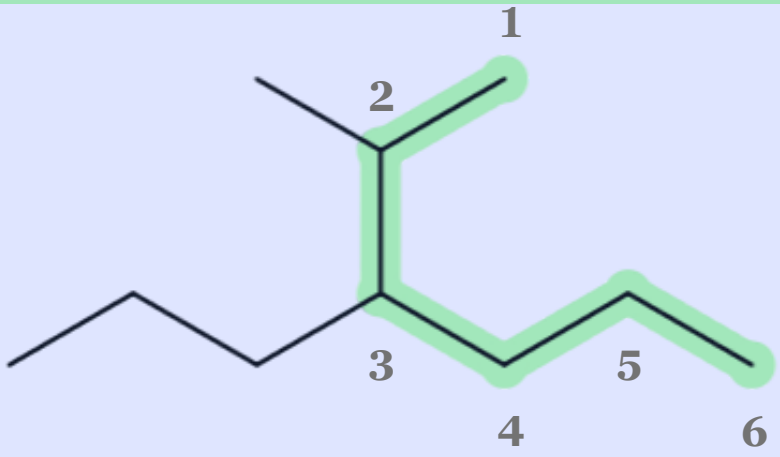
Returned name: 1-ethyl-3-methylcyclohexane
✓ Match

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A09 - STRUCTURE - NAME

CHALLENGE | MOLECULAR FORMULA: C10H22

1 - NUMBERED PARENT



2 - NAMING TRAIL

Parent: longest chain = 6C → **hexane**. There are 4 ways to achieve this chain.

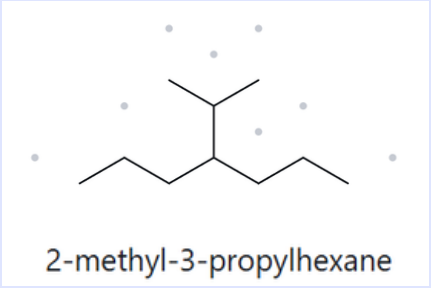
Numbering: from uppermost and going down gives 2 as the lowest locant; from bottom and going up gives 4 as the lowest locant. So start from the uppermost.

Substituents: C2 shows a methyl group and C3 gives propyl group → **2-methyl-3-propyl**

3 - MY NAME (WRITTEN FIRST)

2-methyl-3-propylhexane

4 - THE ALKANE CHECK



Returned name: 2-methyl-3-propylhexane

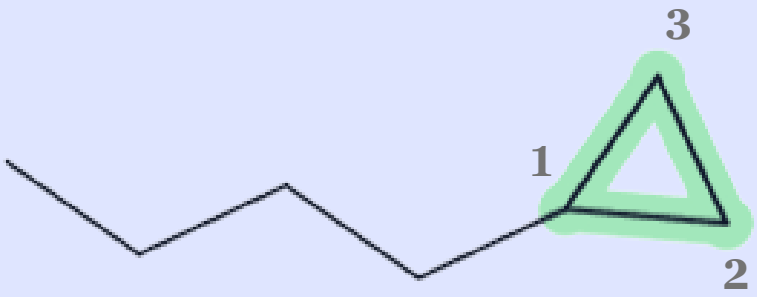
✓ **Match**

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A10 - STRUCTURE - NAME

CHALLENGE | MOLECULAR FORMULA: C7H12

1 - NUMBERED PARENT



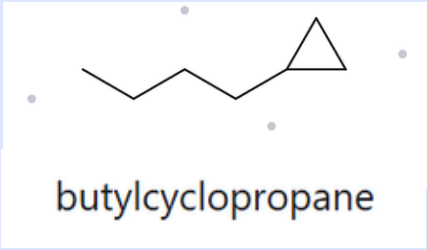
2 - NAMING TRAIL

Parent: ring of 3C is the parent → **cyclopropane**
Numbering: one locant only so not needed
Substituents: butyl group → **butyl**

3 - MY NAME (WRITTEN FIRST)

butylcyclopropane

4 - THE ALKANE CHECK



Returned name: butylcyclopropane
✓ **Match**

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

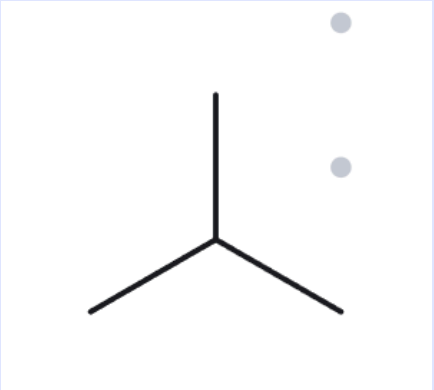
CARD B01 - NAME - STRUCTURE

FOUNDATION | GIVEN NAME: 2-methylpropane

1 - NAME DECODED

Parent: propane → a 3-carbon chain
Substituents: methyl on C2
Expected Formula: 3 + 1 = 4C; open chain → C4H10

2 - BOND-LINE STRUCTURE (THE ALKANE)



3 - FORMULA CHECK

Counted from the drawing: C4H10

CnH2n+2 = C4H10 ✓

4 - BACK-CHECK

Returned by the Alkane:

2-methylpropane

✓ Same as the given name

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

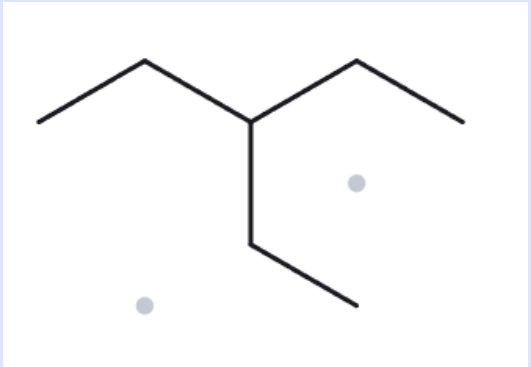
CARD B02 - NAME - STRUCTURE

FOUNDATION | GIVEN NAME: 3-ethylpentane

1 - NAME DECODED

Parent: pentane → a 5-carbon chain
Substituents: ethyl on C3
Expected Formula: 5 + 2 = 7C; open chain → C7H16

2 - BOND-LINE STRUCTURE (THE ALKANE)



3 - FORMULA CHECK

Counted from the drawing: C7H16

CnH2n+2 = C7H16 ✓

4 - BACK-CHECK

Returned by the Alkane:

3-ethylpentane

✓ Same as the given name

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

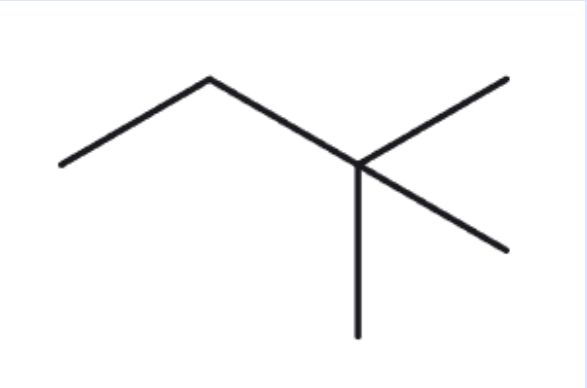
CARD B03 - NAME - STRUCTURE

FOUNDATION | GIVEN NAME: 2,2-dimethylbutane

1 - NAME DECODED

Parent: butane → a 4-carbon chain
Substituents: 2 methyl groups on C2
Expected Formula: 4 + 1(2) = 6C; open chain → C6H14

2 - BOND-LINE STRUCTURE (THE ALKANE)



3 - FORMULA CHECK

Counted from the drawing: C6H14

CnH2n+2 = C6H14 ✓

4 - BACK-CHECK

Returned by the Alkane:

2,2-dimethylbutane

✓ Same as the given name

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

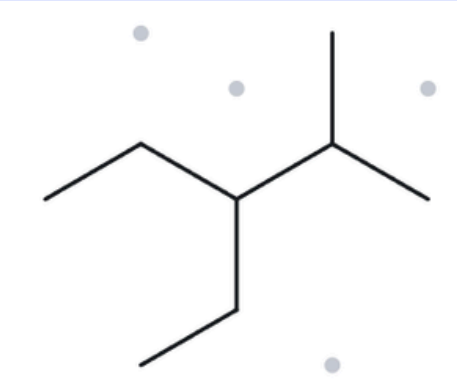
CARD B04 - NAME - STRUCTURE

DEVELOPING | GIVEN NAME: 3-ethyl-2-methylpentane

1 - NAME DECODED

Parent: pentane → a 5-carbon chain
Substituents: methyl on C2 and ethyl on C3
Expected Formula: 5 + 2 + 1 = 8C; open chain → C8H18

2 - BOND-LINE STRUCTURE (THE ALKANE)



3 - FORMULA CHECK

Counted from the drawing: C8H18

CnH2n+2 = C8H18 ✓

4 - BACK-CHECK

Returned by the Alkane:

3-ethyl-2-methylpentane

✓ Same as the given name

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

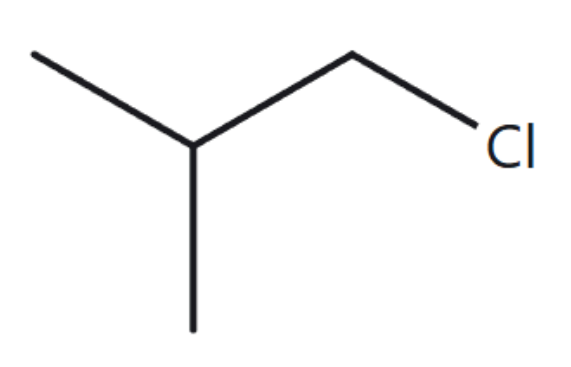
CARD B05 - NAME - STRUCTURE

DEVELOPING | GIVEN NAME: 1-chloro-2-methylpropane

1 - NAME DECODED

Parent: propane → a 3-carbon chain
Substituents: methyl on C2 and chloro on C1
Expected Formula: 3 + 1 = 4C; open chain → C4H10; Cl replaces one H → C4H9Cl

2 - BOND-LINE STRUCTURE (THE ALKANE)



3 - FORMULA CHECK

Counted from the drawing: C4H9Cl

CnH2n+1 Cl = C4H9Cl ✓

4 - BACK-CHECK

Returned by the Alkane:

1-chloro-2-methylpropane

✓ Same as the given name

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

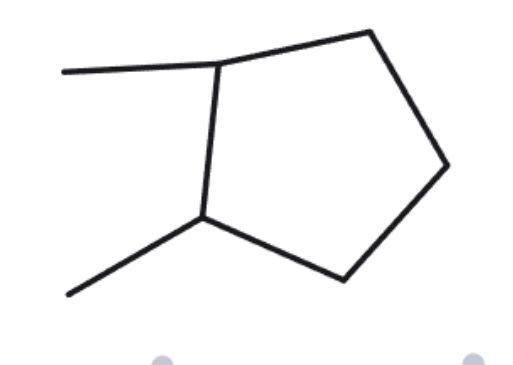
CARD B06 - NAME - STRUCTURE

DEVELOPING | GIVEN NAME: 1,2-dimethylcyclopentane

1 - NAME DECODED

Parent: cyclopentane → a 5-carbon ring
Substituents: 2 methyl groups on C1 and C2
Expected Formula: 5 + 1(2) = 7C; one ring → C7H14

2 - BOND-LINE STRUCTURE (THE ALKANE)



3 - FORMULA CHECK

Counted from the drawing: C7H14

One Ring: CnH2n = C7H14 ✓

4 - BACK-CHECK

Returned by the Alkane:

1,2-dimethylcyclopentane

✓ Same as the given name

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

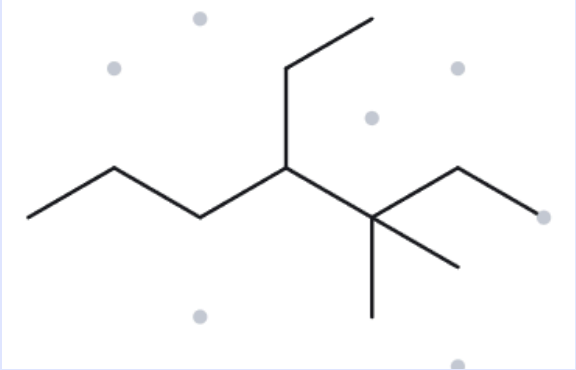
CARD B07 - NAME - STRUCTURE

DEVELOPING | GIVEN NAME: 4-ethyl-3,3-dimethylheptane

1 - NAME DECODED

Parent: heptane → a 7-carbon chain
Substituents: 2 methyls on C3 and ethyl on C4
Expected Formula: 7 + 1(2) + 2 = 11C; open chain → C11H24

2 - BOND-LINE STRUCTURE (THE ALKANE)



3 - FORMULA CHECK

Counted from the drawing: C11H24

CnH2n+1 Cl = C11H24 ✓

4 - BACK-CHECK

Returned by the Alkane:

4-ethyl-3,3-dimethylheptane

✓ Same as the given name

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

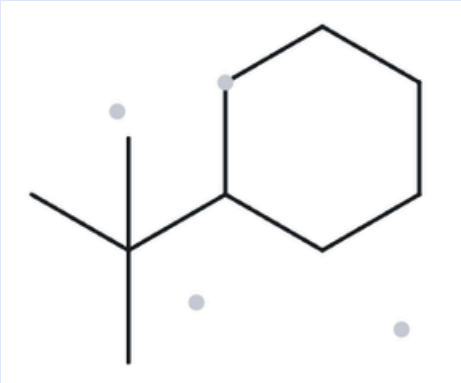
CARD B08 - NAME - STRUCTURE

CHALLENGE | GIVEN NAME: tert-butylcyclohexane

1 - NAME DECODED

Parent: cyclohexane → a 6-carbon ring
Substituents: tert-butyl group on C1
Expected Formula: 6 + 4 = 10C; one ring → C10H20

2 - BOND-LINE STRUCTURE (THE ALKANE)



3 - FORMULA CHECK

Counted from the drawing: C10H20
One Ring: CnH2n = C10H20 ✓

4 - BACK-CHECK

Returned by the Alkane:

(1,1-dimethylethyl)cyclohexane

✗ **NOT the same as the given name**
though tert-butyl and 1,1-dimethyl are equivalent forms

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

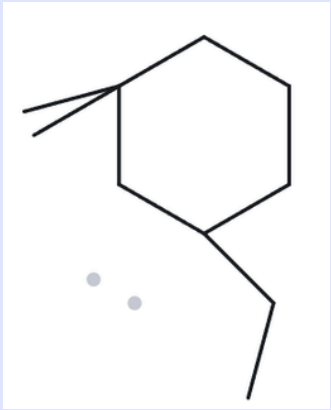
CARD B09 - NAME - STRUCTURE

CHALLENGE | GIVEN NAME: 3-ethyl-1,1-dimethylcyclohexane

1 - NAME DECODED

Parent: cyclohexane → a 6-carbon ring
Substituents: 2 methyl groups on C1 and an ethyl on C3
Expected Formula: 6 + 1(2) + 2 = 10C; one ring → C10H20

2 - BOND-LINE STRUCTURE (THE ALKANE)



3 - FORMULA CHECK

Counted from the drawing: C10H20
One Ring: CnH2n = C10H20 ✓

4 - BACK-CHECK

Returned by the Alkane:

3-ethyl-1,1-dimethylcyclohexane

✓ Same as the given name

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

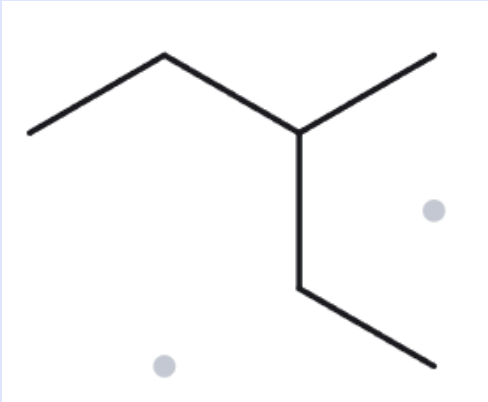
CARD B10 - NAME - STRUCTURE

CHALLENGE | GIVEN NAME: 2-ethylbutane

1 - NAME DECODED

Parent: butane → a 4-carbon chain
Substituents: ethyl on C2
Expected Formula: 4 + 2 = 6C; open chain → C6H14

2 - BOND-LINE STRUCTURE (THE ALKANE)



3 - FORMULA CHECK

Counted from the drawing: C6H14
CnH2n+1 Cl = C6H14 ✓

4 - BACK-CHECK

Returned by the Alkane:

3-methylpentane

✗ **NOT the same as the given name**
It is incorrect because the longest continuous carbon chain actually has five carbons rather than four, making the correct name 3-methylpentane.

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded